ResponseFlow API

API Design Document

1. API Domain Overview

The ResponseFlow API is a backend system for managing customer service requests in real time across hospitality and service businesses such as hotels, restaurants, country clubs, and retail stores.

It tracks guest requests, staff assignments, and service response times, giving managers visibility into operational performance and helping businesses reduce delays and improve accountability.

2. API Resources

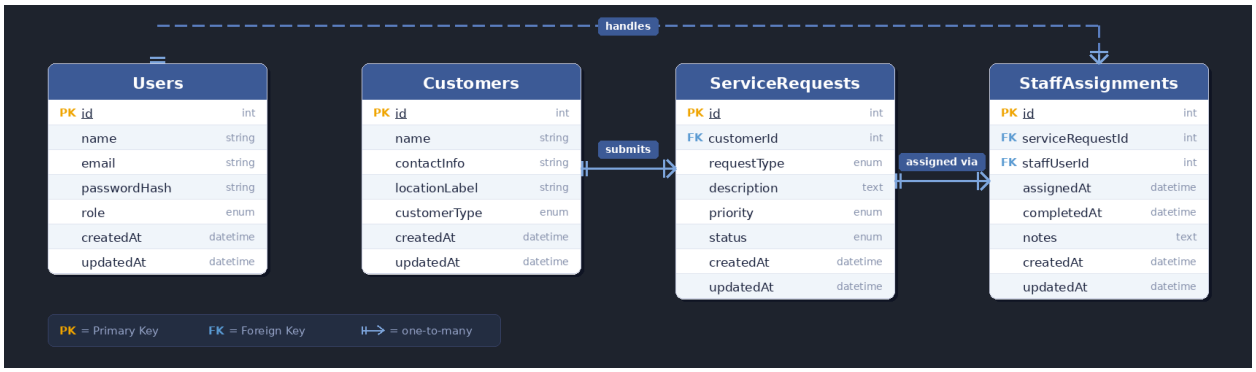
The three main CRUD resources:

- Customers — the people submitting service requests
- ServiceRequests — the core object tracking each request through its lifecycle
- StaffAssignments — connects a service request to the staff member handling it

User resource (authentication/authorization only):

- Users — stores accounts with roles: staff, manager, admin

3. ER Diagram



4. Authentication Endpoints

POST /api/auth/signup

Endpoint	`POST /api/auth/signup`
Description	Register a new user account. Returns a JWT token on success.
Access Control	Public — no token required
Request Body	{ name, email, password, role }
Success Response	201 Created { "token": "<jwt>", "user": { "id": 1, "name": "...", "role": "staff" } }
Error Cases	400 Bad Request — missing or invalid fields 409 Conflict — email already in use

POST /api/auth/login

Endpoint	`POST /api/auth/login`
Description	Log in with email and password. Returns a JWT token on success.
Access Control	Public — no token required
Request Body	{ email, password }
Success Response	200 OK { "token": "<jwt>", "user": { "id": 4, "name": "...", "role": "manager" } }
Error Cases	400 Bad Request — missing fields 401 Unauthorized — invalid email or password

5. Main Resource 1: Customers

POST /api/customers

Endpoint	`POST /api/customers`
Description	Create a new customer record.
Access Control	Manager, Admin
Success Response	201 Created { "id": 1, "name": "John Smith", "customerType": "guest", "locationLabel": "Table 4" }
Error Cases	400 — missing required fields 401 — no token 403 — staff cannot create customers

GET /api/customers

Endpoint	`GET /api/customers`
Description	Retrieve all customers. Optional filter: ?customerType=member
Access Control	Manager, Admin
Success Response	200 OK [{ "id": 1, "name": "John Smith", "customerType": "guest" }, ...]
Error Cases	401 — no token 403 — staff cannot list all customers

GET /api/customers/:id

Endpoint	`GET /api/customers/:id`
Description	Retrieve a single customer by ID.
Access Control	Staff, Manager, Admin
Success Response	200 OK { "id": 1, "name": "John Smith", "contactInfo": "555-1234", "locationLabel": "Table 4" }
Error Cases	400 — ID is not a positive integer 401 — no token 404 — customer not found

PUT /api/customers/:id

Endpoint	`PUT /api/customers/:id`
Description	Update a customer's information.
Access Control	Manager, Admin
Success Response	200 OK { "id": 1, "name": "John Smith", "locationLabel": "Room 12", "updatedAt": "..." }
Error Cases	400 — invalid fields 401 — no token 403 — staff cannot update 404 — not found

DELETE /api/customers/:id

Endpoint	`DELETE /api/customers/:id`
Description	Delete a customer record.
Access Control	Admin only
Success Response	204 No Content
Error Cases	400 — invalid ID 401 — no token 403 — only admin can delete 404 — not found

6. Main Resource 2: ServiceRequests

POST /api/service-requests

Endpoint	`POST /api/service-requests`
Description	Create a new service request for a customer.
Access Control	Manager, Admin
Success Response	201 Created { "id": 17, "customerId": 3, "requestType": "water_refill", "priority": "medium", "status": "open" }
Error Cases	400 — missing customerId or requestType 401 — no token 403 — forbidden 404 — customerId not found

GET /api/service-requests

Endpoint	`GET /api/service-requests`
Description	Retrieve all service requests. Filters: ?status=open, ?priority=high
Access Control	Staff, Manager, Admin
Success Response	200 OK [{ "id": 17, "status": "open", "priority": "high", "requestType": "manager_help" }, ...]
Error Cases	401 — no token

GET /api/service-requests/:id

Endpoint	`GET /api/service-requests/:id`
Description	Retrieve a single service request by ID.
Access Control	Staff, Manager, Admin
Success Response	200 OK { "id": 17, "customerId": 3, "requestType": "water_refill", "status": "assigned", "priority": "medium" }
Error Cases	400 — invalid ID 401 — no token 404 — not found

PUT /api/service-requests/:id

Endpoint	`PUT /api/service-requests/:id`
Description	Update a service request's status or priority.
Access Control	Staff (own assigned only), Manager, Admin
Success Response	200 OK { "id": 17, "status": "in_progress", "updatedAt": "..." }
Error Cases	400 — invalid status transition (e.g. completed without assignment) 401 — no token 403 — forbidden 404 — not found

DELETE /api/service-requests/:id

Endpoint	`DELETE /api/service-requests/:id`
Description	Cancel or delete a service request.
Access Control	Manager, Admin
Success Response	204 No Content
Error Cases	400 — invalid ID 401 — no token 403 — forbidden 404 — not found

7. Main Resource 3: StaffAssignments

POST /api/staff-assignments

Endpoint	`POST /api/staff-assignments`
Description	Assign a service request to a staff member.
Access Control	Manager, Admin
Success Response	201 Created { "id": 5, "serviceRequestId": 17, "staffUserId": 3, "assignedAt": "..." }
Error Cases	400 — missing fields or request is cancelled 401 — no token 403 — forbidden 404 — request or user not found

GET /api/staff-assignments

Endpoint	`GET /api/staff-assignments`
Description	Retrieve all assignments. Filter: ?staffUserId=3
Access Control	Manager, Admin
Success Response	200 OK [{ "id": 5, "serviceRequestId": 17, "staffUserId": 3, "assignedAt": "..." }, ...]
Error Cases	401 — no token 403 — staff cannot list all assignments

GET /api/staff-assignments/:id

Endpoint	`GET /api/staff-assignments/:id`
Description	Retrieve a single assignment by ID.
Access Control	Staff (own only), Manager, Admin
Success Response	200 OK { "id": 5, "serviceRequestId": 17, "staffUserId": 3, "completedAt": null, "notes": "" }
Error Cases	400 — invalid ID 401 — no token 403 — forbidden 404 — not found

PUT /api/staff-assignments/:id

Endpoint	`PUT /api/staff-assignments/:id`
Description	Update notes or mark assignment complete. completedAt is auto-set on completion.
Access Control	Staff (own only), Manager, Admin
Success Response	200 OK { "id": 5, "notes": "Refill done", "completedAt": "2026-03-31T14:35:00Z" }
Error Cases	400 — invalid fields 401 — no token 403 — staff cannot update others 404 — not found

DELETE /api/staff-assignments/:id

Endpoint	`DELETE /api/staff-assignments/:id`
Description	Remove or reassign a staff assignment.
Access Control	Manager, Admin
Success Response	204 No Content
Error Cases	400 — invalid ID 401 — no token 403 — forbidden 404 — not found